Marlos C. Machado
CMPUT 365: Introduction to Reinforcement Learning

Worksheet 4: Dynamic Programming

Question 1. In iterative policy evaluation, we seek to find the value function for a policy π by applying the Bellman equation many times to generate a sequence of value functions v_k that will eventually converge to the true value function v_π. How can we modify the update below to generate a sequence of action value functions q_k?

$$v_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)[r + \gamma v_k(s')]$$

$$q_{k+1}(s,a) = \sum_{s',r} p(s',r|s,a)\Big[r + \gamma \sum_{a'} \pi(a'|s') q_k(s',a')\Big]$$

Question 2. A deterministic policy $\pi(s)$ outputs an action $a \in \mathcal{A} = \{a_1, a_2, \ldots, a_k\}$ directly. More generally, a policy $\pi(\cdot|s)$ outputs the probabilities for all actions: $\pi(\cdot|s) = [\pi(a_1|s), \pi(a_2|s), \ldots, \pi(a_k|s)]$. How can you write a deterministic policy in this form? Let $\pi(s) = a_i$ and define $\pi(\cdot|s)$.

$\pi(\cdot|s)[k] = 1$ if $k = \pi(s)$ and 0 otherwise.

Question 3. Exercise 4.1 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 4. Exercise 4.2 from Sutton & Barto (2018).

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Question 5. [**Challenge**] Exercise 4.4 from Sutton & Barto (2018).

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Question 6. [**Challenge**] A gambler has the opportunity to make bets on the outcomes of a sequence of coin flips. If the coin comes up heads, she wins as many dollars as she has staked on that flip; if it is tails, she loses her stake. The game ends when the gambler wins by reaching her goal of \$100, or loses by running out of money. On each flip, the gambler must decide what portion of her capital to stake, in integer numbers of dollars. This problem can be formulated as an undiscounted, episodic, finite MDP. The state is the gambler's capital, $s \in \{1, 2, \ldots, 99\}$ and the actions are stakes, $a \in \{0, 1, \ldots, \min(s, 100 - s)\}$. The reward is $+1$ when reaching the goal of \$100 and zero on all other transitions. The probability of seeing heads is $p_h = 0.4$.

(a) What does the value of a state mean in this problem? For example, in a gridworld where the value of 1 per step, the value represents the expected number of steps to goal. What does the value of state mean in the gambler's problem? Think about the minimum and maximum possible values, and think about the values of state 50 (which is 0.4) and state 99 (which is near 0.95).

(b) Modify the pseudocode for value iteration to more efficiently solve this specific problem, by exploiting your knowledge of the dynamics. *Hint: Not all states transition to every other state. For example, can you transition from state* 1 *to state* 99*?*

(a) $v)\pi(s) = \mathbb{E}[G_t = s]$ and G_t can have a value of 0 or 1. $G_t = 1$ if the game is won and 0 otherwise. Therefore, the value of a state determines the probability of winning from that state.

(b) Given that action a is taken in state s, the set of possible next states is $\{s - a, s + a\}$. Therefore, we can update $V(s)$ as shown below:

$$V(s) \leftarrow \max_a \sum_{s' \in \{s-a, s+a\}, r} p(s', r|s, a)[r + \gamma v_k(s')]$$

And we can also compute $pi(s)$ as shown below:

$$\pi(s) \leftarrow \arg\max_a \sum_{s' \in \{s-a, s+a\}, r} p(s', r|s, a)[r + \gamma v_k(s')].$$